

ENTERPRISE AVIATION DATA PLATFORM (EADP)

UC8 — Product Backlog | Agile Scrum | Sprint 0-3

Backlog ID	Epic	User Story / Requirement	Description	Priority	Sprint	Story Points	Acceptance Criteria	Dependency	Status	Owner
PB-001	Project Initiation	Study airline business problem	Understand ABC Airways operational and analytics challenges.	High	Sprint 0	3	Business problem, objectives, challenges and outcomes documented.	None	Completed	
PB-002	Project Initiation	Identify business stakeholders	Identify Flight Operations, Airport Operations, Customer Service, Maintenance, Commercial and Executive Management stakeholders.	High	Sprint 0	2	All six stakeholder groups documented with responsibilities.	PB-001	Completed	
PB-003	Project Initiation	Define project objectives	Document objectives for an enterprise-grade aviation data platform.	High	Sprint 0	2	Objectives cover heterogeneous ingestion, quality, warehouse, governance, analytics and Git.	PB-001	Completed	
PB-004	Project Initiation	Define project scope	Define in-scope and out-of-scope project boundaries.	High	Sprint 0	3	Scope is documented and approved for sprint planning.	PB-003	Completed	
PB-005	Project Initiation	Prepare Business Requirements Document	Document business requirements for the EADP project.	High	Sprint 0	5	BRD exists and reflects the project business problem, objectives, stakeholders and scope.	PB-001; PB-002; PB-004	Completed	
PB-006	Project Initiation	Create project charter	Document project purpose, scope, stakeholders, milestones and governance.	High	Sprint 0	3	Project charter is available in docs.	PB-004	Completed	
PB-007	Project Initiation	Prepare Product Backlog	Create prioritized backlog covering project work across all sprints.	High	Sprint 0	3	Backlog has IDs, priorities, sprints, estimates, acceptance criteria and status.	PB-005; PB-006	In Progress	
PB-008	Architecture & Repository	Design enterprise data architecture	Define the end-to-end data platform architecture.	High	Sprint 0	5	Architecture shows source systems, ETL, landing/staging, curated, warehouse and analytics layers.	PB-004	Completed	
PB-009	Architecture & Repository	Design high-level architecture diagram	Create architecture diagram for the EADP platform.	High	Sprint 0	5	Architecture diagram is available and version controlled.	PB-008	Completed	
PB-010	Architecture & Repository	Create Git repository	Initialize local repository and connect it to GitHub.	High	Sprint 0	2	Repository is initialized, remote configured and main branch pushed.	PB-004	Completed	
PB-011	Architecture & Repository	Create enterprise repository structure	Create docs, datasets, bronze, silver, gold, pentaho, python, sql, metadata, lineage, dashboards, deployment and tests directories.	High	Sprint 0	3	Required directory structure exists in repository.	PB-010	Completed	
PB-012	Architecture & Repository	Define Git commit convention	Define meaningful conventional commit prefixes for project history.	Medium	Sprint 0	2	Commit convention documented and followed.	PB-010	Completed	
PB-013	Architecture & Repository	Define sprint milestone tagging strategy	Define sprint-0, sprint-1 and final-release tags.	Medium	Sprint 0	2	Tagging strategy documented and used at sprint milestones.	PB-010	Not Started	
PB-014	Source Discovery	Identify Flight Operations data source	Catalog Flight Operations source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-015	Source Discovery	Identify Passenger Reservation data source	Catalog Passenger Reservation System source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-016	Source Discovery	Identify Airport Operations data source	Catalog Airport Operations source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-017	Source Discovery	Identify Baggage Handling data source	Catalog Baggage Handling source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-018	Source Discovery	Identify Aircraft Maintenance data source	Catalog Aircraft Maintenance source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-019	Source Discovery	Identify Crew Scheduling data source	Catalog Crew Scheduling source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-020	Source Discovery	Identify Ticketing and Booking data source	Catalog Ticketing and Booking source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-021	Source Discovery	Identify CRM data source	Catalog CRM source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-022	Source Discovery	Identify Flight Delay data source	Catalog Flight Delay source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-023	Source Discovery	Identify Aircraft IoT data source	Catalog Aircraft IoT source data.	High	Sprint 1	2	Source, format, entity, owner, ingestion method and target are recorded.	PB-007	Completed	
PB-024	Source Discovery	Prepare Source Inventory	Create a consolidated source inventory for all identified airline systems.	High	Sprint 1	5	All identified sources include format, business area, ingestion method, target layer/table and ownership.	PB-014; PB-015; PB-016; PB-017; PB-018; PB-019; PB-020; PB-021; PB-022; PB-023	Not Started	
PB-025	Source Discovery	Prepare Data Dictionary	Define source fields, types, business definitions, nullability, keys, targets and transformation rules.	High	Sprint 1	5	Dictionary covers source and target fields for ingested datasets.	PB-024	Not Started	
PB-026	Data Ingestion	Design ingestion framework	Define common ingestion pattern for CSV, Excel, JSON, XML and SQL sources.	High	Sprint 1	3	Framework defines extract, validate, transform, load, logging and exception handling.	PB-024; PB-025	Not Started	
PB-027	Data Ingestion	Implement CSV ingestion	Build Pentaho transformations for CSV datasets.	High	Sprint 1	5	CSV sources load successfully into PostgreSQL staging with mapping and validation.	PB-026	Not Started	
PB-028	Data Ingestion	Implement Excel ingestion	Build Pentaho transformations for Excel datasets.	High	Sprint 1	5	Excel sources load successfully into PostgreSQL staging with mapping and validation.	PB-026	Not Started	
PB-029	Data Ingestion	Implement JSON ingestion	Build Pentaho transformations for JSON datasets.	High	Sprint 1	5	JSON sources load successfully into PostgreSQL staging with mapping and validation.	PB-026	Not Started	
PB-030	Data Ingestion	Implement XML ingestion	Build Pentaho transformations for XML data.	High	Sprint 1	5	XML source loads successfully into PostgreSQL staging with mapping and validation.	PB-026	Not Started	
PB-031	Data Ingestion	Implement SQL ingestion	Build Pentaho transformation for SQL/relational source data.	High	Sprint 1	5	SQL source is extracted and loaded into PostgreSQL staging with validation.	PB-026	Not Started	
PB-032	Data Ingestion	Implement PostgreSQL staging tables	Create staging schema and source-aligned staging tables.	High	Sprint 1	5	Staging tables exist with correct data types, keys where applicable and audit columns as designed.	PB-025	Not Started	
PB-033	Data Ingestion	Implement ingestion logging	Capture pipeline start/end, source, records read, loaded, rejected, errors and status.	High	Sprint 1	3	Successful and failed pipeline executions generate useful operational logs.	PB-026	Not Started	
PB-034	Data Ingestion	Implement exception handling	Handle invalid records and pipeline failures without silently losing errors.	High	Sprint 1	3	Exceptions are captured, rejected records are traceable and pipeline status is visible.	PB-026	Not Started	
PB-035	Data Ingestion	Validate ingested records	Perform row count, null, duplicate, datatype and key validation after ingestion.	High	Sprint 1	5	Validation checks pass or failures are documented and traceable.	PB-027; PB-028; PB-029; PB-030; PB-031; PB-032	Not Started	
PB-036	Data Profiling & Quality	Profile aviation datasets using Python/Pandas	Generate structural and statistical profiles of source/staging datasets.	High	Sprint 2	5	Profiles include schema, missingness, distributions and basic quality indicators.	PB-035	Not Started	
PB-037	Data Profiling & Quality	Analyze missing values	Identify and report missing values in critical fields.	High	Sprint 2	3	Missingness is quantified by dataset and critical field.	PB-036	Not Started	
PB-038	Data Profiling & Quality	Detect duplicate passenger records	Identify duplicate passenger records and document rules.	High	Sprint 2	3	Duplicate candidates are identified and remediation rules documented.	PB-036	Not Started	
PB-039	Data Profiling & Quality	Identify flight data inconsistencies	Detect invalid or inconsistent flight information.	High	Sprint 2	3	Flight quality exceptions are identified and documented.	PB-036	Not Started	
PB-040	Data Profiling & Quality	Analyze baggage data quality	Profile baggage records for missing, duplicate and inconsistent values.	Medium	Sprint 2	3	Baggage quality findings are documented.	PB-036	Not Started	
PB-041	Data Profiling & Quality	Analyze aircraft maintenance data quality	Profile maintenance records for data quality issues.	High	Sprint 2	3	Maintenance quality findings are documented.	PB-036	Not Started	
PB-042	Data Profiling & Quality	Standardize master data	Standardize passenger, airport, aircraft and flight master data.	High	Sprint 2	5	Standardization rules are applied consistently and exceptions are tracked.	PB-037; PB-038; PB-039; PB-041	Not Started	
PB-043	Data Profiling & Quality	Create data quality report	Summarize profiling results, issues, rules and remediation status.	High	Sprint 2	5	Report includes quality dimensions, findings and recommended actions.	PB-036; PB-037; PB-038; PB-039; PB-040; PB-041	Not Started	
PB-044	Data Warehouse	Design dimensional model	Design star schema for aviation analytics.	High	Sprint 2	5	Fact and dimension relationships support required operational analytics.	PB-042	Not Started	
PB-045	Data Warehouse	Design fact tables	Define measurable business events and grain for fact tables.	High	Sprint 2	5	Fact table grains, measures and foreign keys are documented.	PB-044	Not Started	
PB-046	Data Warehouse	Design dimension tables	Define passenger, airport, aircraft, flight and other required dimensions.	High	Sprint 2	5	Dimension attributes, keys and relationships are documented.	PB-044	Not Started	
PB-047	Data Warehouse	Create PostgreSQL warehouse schema	Implement warehouse schemas and tables in PostgreSQL.	High	Sprint 2	5	Warehouse objects are created successfully in PostgreSQL.	PB-045; PB-046	Not Started	
PB-048	Data Warehouse	Load dimension tables	Populate curated dimension tables.	High	Sprint 2	5	Dimensions load successfully with valid keys and standardized attributes.	PB-047	Not Started	
PB-049	Data Warehouse	Load fact tables	Populate aviation fact tables from curated data.	High	Sprint 2	5	Facts load at the defined grain and pass referential checks.	PB-047; PB-048	Not Started	

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PB-050	Data Warehouse	Implement warehouse validation	Validate row counts, referential integrity and business rules.	High	Sprint 2	3	Warehouse validation results are documented.	PB-048; PB-049	Not Started	
PB-051	Data Warehouse	Create operational SQL queries	Develop SQL queries for operational reporting.	Medium	Sprint 2	3	Queries return validated KPIs and operational insights.	PB-049	Not Started	
PB-052	Governance & Lineage	Create metadata repository	Maintain technical and business metadata for platform assets.	High	Sprint 3	5	Metadata repository captures source, target, ownership and transformation information.	PB-025; PB-047	Not Started	
PB-053	Governance & Lineage	Create business glossary	Define common aviation business terms and KPI terminology.	Medium	Sprint 3	3	Business terms have clear definitions and ownership.	PB-052	Not Started	
PB-054	Governance & Lineage	Create source-to-target mapping	Document field-level mappings and transformation rules.	High	Sprint 3	5	Mappings cover source, staging, curated and warehouse targets.	PB-025; PB-047	Not Started	
PB-055	Governance & Lineage	Document data lineage	Document end-to-end source-to-analytics lineage.	High	Sprint 3	5	Lineage identifies movement and transformations from source to reporting.	PB-054	Not Started	
PB-056	Governance & Lineage	Implement data reconciliation	Reconcile passenger, baggage, flight and maintenance data across layers.	High	Sprint 3	5	Reconciliation checks are executed and exceptions documented.	PB-050; PB-055	Not Started	
PB-057	Governance & Lineage	Document data quality rules	Document validation and quality rules used across the platform.	Medium	Sprint 3	3	Rules are traceable to datasets and validation processes.	PB-043	Not Started	
PB-058	ETL Orchestration	Create Pentaho transformation structure	Organize .ktr transformations by source and purpose.	High	Sprint 1	3	Transformation directory is organized and naming convention is followed.	PB-026	Not Started	
PB-059	ETL Orchestration	Create Pentaho job workflows	Orchestrate transformations using Pentaho jobs.	High	Sprint 3	5	Jobs execute required dependencies in the correct sequence.	PB-033; PB-034	Not Started	
PB-060	ETL Orchestration	Implement job-level logging	Capture execution status and operational metrics for jobs.	High	Sprint 3	3	Job runs provide traceable execution logs.	PB-059	Not Started	
PB-061	ETL Orchestration	Implement failure handling	Implement job-level failure and recovery handling.	High	Sprint 3	3	Failures are detected, logged and routed for investigation.	PB-059; PB-060	Not Started	
PB-062	ETL Orchestration	Test complete ETL workflow	Perform end-to-end ETL integration testing.	High	Sprint 3	5	Complete workflow executes successfully and validation results are documented.	PB-059; PB-061	Not Started	
PB-063	Analytics & Dashboards	Define aviation KPIs	Define KPIs for flight, passenger, maintenance, baggage and executive analytics.	High	Sprint 3	3	KPI definitions, formulas, dimensions and data sources are documented.	PB-051; PB-053	Not Started	
PB-064	Analytics & Dashboards	Create flight operations dashboard	Develop dashboard for flight punctuality and delay trends.	High	Sprint 3	5	Dashboard supports flight punctuality and delay analysis.	PB-063	Not Started	
PB-065	Analytics & Dashboards	Create passenger analytics dashboard	Develop dashboard for passenger booking and travel patterns.	High	Sprint 3	5	Dashboard supports passenger booking/travel analysis.	PB-063	Not Started	
PB-066	Analytics & Dashboards	Create aircraft and maintenance dashboard	Develop dashboard for aircraft utilization and maintenance schedules.	High	Sprint 3	5	Dashboard supports utilization and maintenance monitoring.	PB-063	Not Started	
PB-067	Analytics & Dashboards	Create baggage analytics dashboard	Develop dashboard for baggage handling performance.	Medium	Sprint 3	5	Dashboard supports baggage performance analysis.	PB-063	Not Started	
PB-068	Analytics & Dashboards	Create executive dashboard	Develop executive operational and financial reporting dashboard.	High	Sprint 3	5	Dashboard provides executive-level operational and financial insights.	PB-063	Not Started	
PB-069	Analytics & Dashboards	Validate dashboard metrics	Reconcile dashboard KPIs to trusted warehouse queries.	High	Sprint 3	3	Dashboard metrics match validated source/warehouse calculations.	PB-064; PB-065; PB-066; PB-067; PB-068	Not Started	
PB-070	Testing & Documentation	Create ingestion test cases	Document test cases for source ingestion pipelines.	High	Sprint 1	3	Test cases cover successful, invalid and exception scenarios.	PB-035	Not Started	
PB-071	Testing & Documentation	Create data quality test cases	Document test cases for profiling and quality rules.	High	Sprint 2	3	Test cases cover missing, duplicate, consistency and validity checks.	PB-043	Not Started	
PB-072	Testing & Documentation	Perform integration testing	Test interaction between ingestion, staging, curated and warehouse components.	High	Sprint 3	5	Integration tests execute with documented results.	PB-050; PB-062	Not Started	
PB-073	Testing & Documentation	Perform end-to-end testing	Validate the full source-to-dashboard workflow.	High	Sprint 3	5	End-to-end flow passes critical acceptance tests.	PB-069; PB-072	Not Started	
PB-074	Testing & Documentation	Prepare final technical documentation	Compile architecture, ETL, database, governance, testing and analytics documentation.	High	Sprint 3	5	Final technical documentation is complete and version controlled.	PB-073	Not Started	
PB-075	Testing & Documentation	Prepare final project presentation	Prepare final presentation of solution, results and learnings.	High	Sprint 3	3	Presentation covers problem, architecture, implementation, analytics and outcomes.	PB-074	Not Started	
PB-076	Testing & Documentation	Create final Git release	Tag and document final project release in Git.	High	Sprint 3	2	Final release tag exists and repository history is complete.	PB-074; PB-075	Not Started	